Elektronik II

Series-Series, Shunt-Shunt Feedback and Worked Examples

Original notes: Ders12.pdf

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orig. p. 1

Series (o) – Series (I) Feedback Topology

- Suitable for transconductance Amplifier

Figure: Block diagram: source V_s with series R_s feeds the "Basic Transconductance Amplifier" block; the output current I_o (arrow pointing toward the amplifier) flows through the load R_L. The "FB" block sits in series with the output loop (carrying I_o) and its output voltage V_f (– on the source side, + on the amplifier side) is in series with the input loop.
(in the original notes, p. 1)

Figure: Complete model: source V_s with terminals s, s' feeds R_i (voltage v_i, + top, – bottom). Output side: dependent current source Av_i in parallel with R_o, output terminals o, o'. The dependent voltage source βI_o (+ top) sits in the input loop between s' and the bottom of R_i, and the output current path loops back through the feedback section. Red dashed boxes: "A circuit" (R_i, Av_i, R_o) and "β circuit" (βI_o).
(in the original notes, p. 1)

Two-port equivalent circuit

Figure: Two-port equivalent: source V_i with R_s (top wire) and R_{11} (bottom wire) feeds the box "BTA"; at the output the current I_o (arrow toward the box) flows through R_L and R_{22} (bottom wire); an arrow labelled R_o looks in at the output terminals.
(in the original notes, p. 1)

Figure: "FN" box with open terminals on both sides: arrow R_{11} looking into the input terminals and arrow R_{22} looking into the output terminals.
(in the original notes, p. 1)

$$\frac{I_o}{V_1} = A$$

Figure: "FN" box with "$I_1 = 0$" marked at the input, V_f (+, −) across the input terminals and a current source I_o at the output.
(in the original notes, p. 1)

$$\beta = \frac{V_f}{I_o}$$

$$R_{if} = R_i (1 + A\beta)$$
$$R_{of} = R_o (1 + A\beta)$$
$$A_f = \frac{A}{1 + A\beta}$$

orig. p. 2

Shunt – Shunt feedback topology (Transresistance amplifier)

Figure: Block diagram: current source I_s with parallel R_s feeds the "Base TRA" block; at the output the load R_L with $+v_o-$; the "FBN" block takes the output ($+v_o-$ drawn at its output side, current I_o arrow) and returns the current I_f (arrow) to the input node, in parallel with the source.
(in the original notes, p. 2)

$$A_o = \frac{V_{out}}{I_{in}}$$

Figure: Complete model: current source I_s with terminals s, s' feeds $\mathcal{R}_i$ (voltage v_i, + top, − bottom). Output side: dependent voltage source AI_i in series with $\mathcal{R}_o$, output terminals o, o'. Feedback: dependent current source βV_o (arrow pointing down) connected across the input nodes, controlled by V_o (drawn as $+V_o-$ at the output side). Red dashed box: "A circuit" ($\mathcal{R}_i$, AI_i, $\mathcal{R}_o$).
(in the original notes, p. 2)

Figure: Two reduced circuits. Left: terminals s, s' with the current I_s (arrow) through $\mathcal{R}_{if}$. Right: dependent voltage source $A_f I_s$ in series with $\mathcal{R}_{of}$ to the output terminals, with $+v_o-$.
(in the original notes, p. 2)

Close loop equivalent circuit model

Figure: Equivalent circuit: current source I_i with $\mathcal{R}_s$ and $\mathcal{R}_{11}$ in parallel (arrow labelled $\mathcal{R}_i$ looking into the input terminals) feeds the box "BTRA"; at the output $\mathcal{R}_{22}$ and $\mathcal{R}_L$ are in parallel across the output terminals (+, −, v_o); arrow labelled $\mathcal{R}_o$ at the output.
(in the original notes, p. 2)

orig. p. 3

Figure: Two sketches of the "FN" box. Left: input terminals open with arrow $\mathcal{R}_{11}$ looking into the input, output terminals shorted together. Right: input terminals shorted together, output terminals open with arrow $\mathcal{R}_{22}$ looking into the output.
(in the original notes, p. 3)

Figure: "FN" box with its input shorted and the current I_f (arrow) flowing in the shorted input loop; a voltage source V_o is connected at the output.
(in the original notes, p. 3)

$$\beta = \frac{I_f}{V_o}$$

$$\mathcal{R}_{if} = \frac{\mathcal{R}_i}{1 + A\beta}$$

$$\mathcal{R}_{of} = \frac{\mathcal{R}_o}{1 + A\beta}$$

$$A_f = \frac{A}{1 + A\beta}$$

Example 2

Analyze the circuit below and calculate $\frac{v_o}{v_s}$, R_{in}, and R_{out}

> *Figure:* Circuit: npn transistor with a 4,7k collector resistor to +12 V; a 47k feedback resistor connects the collector to the base; the source v_s (circle with a sine symbol) drives the base through 10k; emitter grounded; 0,7 (+ at the base, – at the emitter) marked across the base-emitter junction; arrow R_{in} looking into the base node.
> *(in the original notes, p. 3)*

ignore r_o! ($V_A = \infty$) $\beta = 100$

> *Figure:* Same circuit redrawn for the solution: current I_y (arrow down) in the 47k feedback resistor, current I_x (arrow down) in the 4,7k collector resistor (to +12 V), −0,07 mA marked on the 10k source resistor, 0,7 across the base-emitter junction, arrow R_{in}.
> *(in the original notes, p. 3)*

$$I_x = \frac{12 - V_C}{4{,}7\,k\Omega} \qquad I_B = I_y - 0{,}07\,mA$$

$$I_C = 100\,I_B \Rightarrow 100\,I_y - 7\,mA = I_C$$

$$I_y = \frac{V_C - 0{,}7}{47k} \qquad I_C = I_x - I_y = \frac{12 - V_C}{4{,}7k} - \left(\frac{V_C - 0{,}7}{47k}\right)$$

$$100\left(\frac{V_C - 0{,}7}{(47k)}\right) - 7\,mA = \frac{12 - V_C}{4{,}7k} - \left(\frac{V_C - 0{,}7}{47k}\right)$$

orig. p. 4

$$V_C = 4{,}95\,V$$

$$I_B = 0{,}015\,mA \qquad g_m = \frac{I_C}{V_T} = 60\,mS$$

$$I_C = 1{,}5\,mA \qquad r_\pi = (\beta + 1)\cdot\frac{1}{g_m} = 1{,}66\,k\Omega$$

a) Feedback type: Shunt (o) – Shunt (I) (Transresistance)

b) Feedback Network:

> *Figure:* Feedback network "FN": a single 47kΩ resistor (dashed outline) between the two terminals.
> *(in the original notes, p. 4)*

c) Calculate R_{11}, R_{22}, and β



Figure: Two sketches of the feedback network (a 47kΩ resistor in a dashed box). Left: the output terminal is joined to the lower input terminal by an orange wire (output shorted); orange arrow $\mathcal{R}_{11}$ looking into the input, labelled $\mathcal{R}_{11} = 47\text{k}\Omega$. Right: resistor with the input side shorted; arrow $\mathcal{R}_{22}$ looking into the output, labelled $\mathcal{R}_{22} = 47\text{k}\Omega$.
(in the original notes, p. 4)



Figure: Circuit: 47kΩ resistor from a node with the current I_f (arrow, node connected to ground) to a voltage source V_o whose other side is grounded.
(in the original notes, p. 4)

$$\frac{I_f}{V_o} = -\frac{1}{47\text{k}\Omega} = \beta$$

d)



Figure: Two-port equivalent: source v_s (sine symbol) with 10kΩ in series, then 47kΩ (marked $\mathcal{R}_{11}$ in orange) and 1,66k in parallel to ground with $v_{\pi 1}$ (+, −) across the latter; output: dependent current source $g_{m1}v_{\pi 1}$ (arrow down) in parallel with 47kΩ (marked $\mathcal{R}_{22}$ in orange) and 4,7kΩ; output voltage v_o and an arrow $\mathcal{R}_o$ looking in from the output.
(in the original notes, p. 4)



Figure: The same circuit with the source replaced by a current source I_i: I_i in parallel with 10kΩ, 47kΩ and 1,66k ($v_{\pi 1}$ across it); output: $g_{m1}v_{\pi 1}$ in parallel with the two output resistors (47kΩ and 4,7kΩ); v_o and arrow $\mathcal{R}_o$ at the output.
(in the original notes, p. 4)

$$A = \frac{v_o}{I_i} \Rightarrow \begin{vmatrix} v_o = -\overset{60\,\text{mS}}{g_{m1}}\, v_{\pi 1}\,(4{,}27\text{k}) \\ v_o = -256{,}2\, v_{\pi 1} \\ v_{\pi 1} = I_i\,(10\text{k} \parallel 47\text{k} \parallel 1{,}66\text{k}) \\ v_{\pi 1} = I_i \cdot 1{,}38\text{k} \end{vmatrix}$$

orig. p. 5

$$v_o = -256{,}2 \cdot I_i \cdot 1{,}38\text{k}$$

$$A = \frac{v_o}{I_i} = -353{,}6\,\text{k}\Omega$$

$$R_i = 1{,}38k$$
$$R_o = 4{,}27k$$

$$A_f = \frac{A}{1 + A\beta} = \frac{-353{,}6k}{1 + 353{,}6k \cdot \frac{1}{47k}} = \frac{-353{,}6\,k\Omega}{8{,}52}$$

$$A_f = -41{,}4\,k\Omega$$

$$R_{if} = \frac{1{,}38k}{8{,}52} \Rightarrow R_{if} = 162\,\Omega$$
$$R_{of} = \frac{4{,}27}{8{,}52} = 501{,}2\,\Omega$$
$$R_{in} = \left(\frac{1}{162} - \frac{1}{10k}\right)^{-1} \approx 165\,\Omega$$
$$R_{out} = R_{of}$$

$$\frac{v_o}{I_i} = -353{,}6\,k\Omega$$
$$\frac{v_o}{v_s} = \frac{A_f}{R_s} = -4{,}14\ [V/V]$$

Example

Analyze the current amplifier given below and calculate: $\frac{v_o}{v_{in}}$, $\frac{I_{out}}{I_{in}}$, R_{in} and R_{out}

Assume $\beta = 100$ for small signal, ∞ for DC

orig. p. 6

$V_A = 75\,V$

Figure: Two-stage amplifier with feedback: source v_s with $R_s = 10k\Omega$ in series feeds (through a coupling capacitor) the base of Q_1, biased by $100k\Omega$ (to +12 V) and $15k\Omega$ (to ground); Q_1 has a $10k\Omega$ collector resistor (to +12 V) and an $870\,\Omega$ emitter resistor; the Q_1 collector drives the base of Q_2, which has an $8k\Omega$ collector resistor (to +12 V) and a $3{,}4k\Omega$ emitter resistor; the Q_2 collector goes through a coupling capacitor (∞) to the load $R_L = 1k\Omega$ (current I_{out}). The feedback resistor $R_f = 10k\Omega$ runs from the Q_2 emitter (through a capacitor, ∞) back to the input node. Orange annotations: "Shunt" at the input node, "Series" at the Q_2 emitter capacitor, and current marks ("I") at the input node and at the Q_2 emitter. *(in the original notes, p. 6)*

Figure: The same circuit annotated with DC values in orange: 1,57 at the base of Q_1, 0,7 across its base-emitter junction, 0,87 at its emitter (the 870 Ω emitter resistor is bypassed by a capacitor, ∞), 1 mA in the 10kΩ collector resistor, 2 V at the Q_1 collector, 0,7 across the base-emitter junction of Q_2, 1,3 V at its emitter, 0,38 (mA) in the 8kΩ resistor and 0,38 mA in the 3,4kΩ emitter resistor.
(in the original notes, p. 6)

$$I_{C1} = 1\,\text{mA} \qquad I_{C2} = 0{,}38\,\text{mA} \qquad r_o = \frac{V_A}{I_C}$$

$$g_{m1} = 40\,\text{mS} \qquad g_{m2} = 15{,}2\,\text{mS}$$

$$r_{\pi 1} = 2{,}5\,\text{k}\Omega \qquad r_{\pi 2} \approx 6{,}6\,\text{k}\Omega$$

$$r_{o1} = 75\,\text{k}\Omega \qquad r_{o2} = \frac{75}{0{,}38} \approx 197\,\text{k}\Omega$$

a) Feedback type : Series-Shunt

b) Feedback network out two port parameters

Figure: Feedback network: 10kΩ in the top wire between the top input and output terminals and 3,4kΩ from the output top node to the bottom wire (ground); output terminals open; arrow $\mathcal{R}_{11}$ looking into the input terminals.
(in the original notes, p. 6)

$$\mathcal{R}_{11} = 13{,}4\,\text{k}\Omega$$

Figure: The same network with the input side shorted (left ends joined top to bottom), so 10kΩ and 3,4kΩ appear in parallel at the output; arrow $\mathcal{R}_{22}$ looking into the output terminals.
(in the original notes, p. 6)

$$\mathcal{R}_{22} = 10\text{k} \parallel 3{,}4\text{k}\Omega \Rightarrow \mathcal{R}_{22} = 2{,}5\,\text{k}\Omega$$

orig. p. 7

$$\beta = \frac{I_f}{I_o} = \frac{-3{,}4\text{k}}{13{,}4\text{k}} \approx -0{,}25\ [\text{A/A}]$$

Figure: Circuit for β: current I_f (arrow) flows through 10k in the top wire; 3,4k in parallel with a current source I_o (arrow up) at the right; the left end returns through a wire along the bottom.
(in the original notes, p. 7)

Figure: Small-signal equivalent circuit: current source $I_s = v_s/R_s$ in parallel with 10kΩ, 13,4kΩ (red), 13kΩ and $r_{\pi 1}$ with $v_{\pi 1}$ (+, −) across it. Then $g_{m1}v_{\pi 1}$ (arrow down) in parallel with $r_{o1} = 75k$, 10kΩ and 6,6k ($r_{\pi 2}$, voltage v_{be2} marked in orange). Then $g_{m2}v_{\pi 2}$ in parallel with $r_{o2} = 197k\Omega$, 8kΩ and the load 1kΩ (output current I_{out}, arrow). A 2,5kΩ resistor in series at the bottom; the output nodes y, y′ are marked with open circles and orange arrows R_o and I_{out}.
(in the original notes, p. 7)

$$A = \frac{I_{out}}{I_s}$$

$$v_{\pi 1} = I_s\,(10k \parallel 13{,}4k \parallel 13k \parallel 2{,}5k) = 1{,}55k\,I_s$$

$$v_{be2} = -g_{m1}\,v_{\pi 1}\left(75k \parallel 10k \parallel (6{,}6k + \underbrace{250k}_{(\beta+1)(2{,}5k)})\right) \approx -320\,v_{\pi 1}$$

$$I_{out} \approx \frac{v_{be2}}{\underbrace{\frac{6{,}6k\Omega}{100} + 2{,}5k}_{2{,}57k\Omega}} \cdot \frac{8}{9}$$

$$\frac{I_{out}}{I_s} = -192{,}4\ [A/A]$$

$$R_i = 1{,}55\,k\Omega$$

$$R_o = 2{,}5k\Omega + \underbrace{\frac{6{,}6k\Omega}{\beta+1}}_{101} + \underbrace{\frac{10k \parallel 75k}{\beta+1}}_{101}$$

$$= 2{,}7\,k\Omega$$

$$A_f = \frac{A}{1 + A\beta}$$

$$1 + A\beta = 1 + (-0{,}25 \times -192{,}4)$$

$$\boxed{1 + A\beta = 49{,}1}$$

$$A_f = \frac{-192{,}4}{49{,}1} \approx -3{,}92\ [A/A]$$

orig. p. 8

$$R_{if} = \frac{R_i}{1 + A\beta} = \frac{1{,}55k}{49{,}1} \approx 31{,}5\,\Omega$$

$$R_{of} = R_o\,(1 + A\beta) = 2{,}7k \times 49{,}1 \approx 132\,k\Omega$$